Product Requirements Document (PRD)

> **Project:** EV-Tech Academy (EV Maintenance Training School)
> **Milestone Target:** 45% MVP Completion & Team Onboarding
> **Document Author:** Technical Product Lead

1. Context & Overview

The traditional automotive repair sector is facing a severe operational bottleneck. As modern electric vehicles dominate the market, mechanics who spent decades mastering internal combustion engines find themselves unprepared for high-voltage architecture. Working blindly around 400V–800V battery packs and precision inverters creates extreme safety hazards. EV-Tech Academy bridges this gap by offering a structured blended learning environment—merging rigorous digital compliance exams with hands-on physical workshop training.

2. Target Audience Profiles

- **Independent Technicians:** Experienced mechanics looking for straightforward, credible high-voltage safety credentials without leaving the shop floor.
- **Garage Owners:** Workshop managers aiming to transition their business into certified EV service centers while mitigating insurance and liability risks.
- **Training Instructors:** Academy supervisors monitoring student test scores, safety pass rates, and physical laboratory capacity.

3. Core MVP Feature Breakdown

Module Name	Functional Description	Technical Delivery
Course Catalog	Presents curated training tracks covering Safety & Compliance, Battery Diagnostics, and Drive Powertrains.	Fetched dynamically via Express REST endpoint (`GET /api/v1/courses`).
Safety Assessment	Mandatory 5-question multiple choice test evaluating NFPA 70E compliance and CAT III/IV tool handling.	Evaluated server-side; issues unique digital certificate ID upon scoring 80%+.
Lab Reservation	Enables mechanics to book physical bench time at regional facilities (Detroit, San Jose, Austin).	Stores reservation payload via `POST /api/v1/bookings` with instant verification.

4. Technical Stack & Architecture

The application is split cleanly into a decoupled architecture:

- **Frontend:** React 18 single-page application built with Vite and styled using Tailwind CSS for an optimal dark-mode technical experience.
- **Backend:** Lightweight Node.js and Express server running on port 5001, providing RESTful endpoints for courses, quiz scoring, and lab bookings.